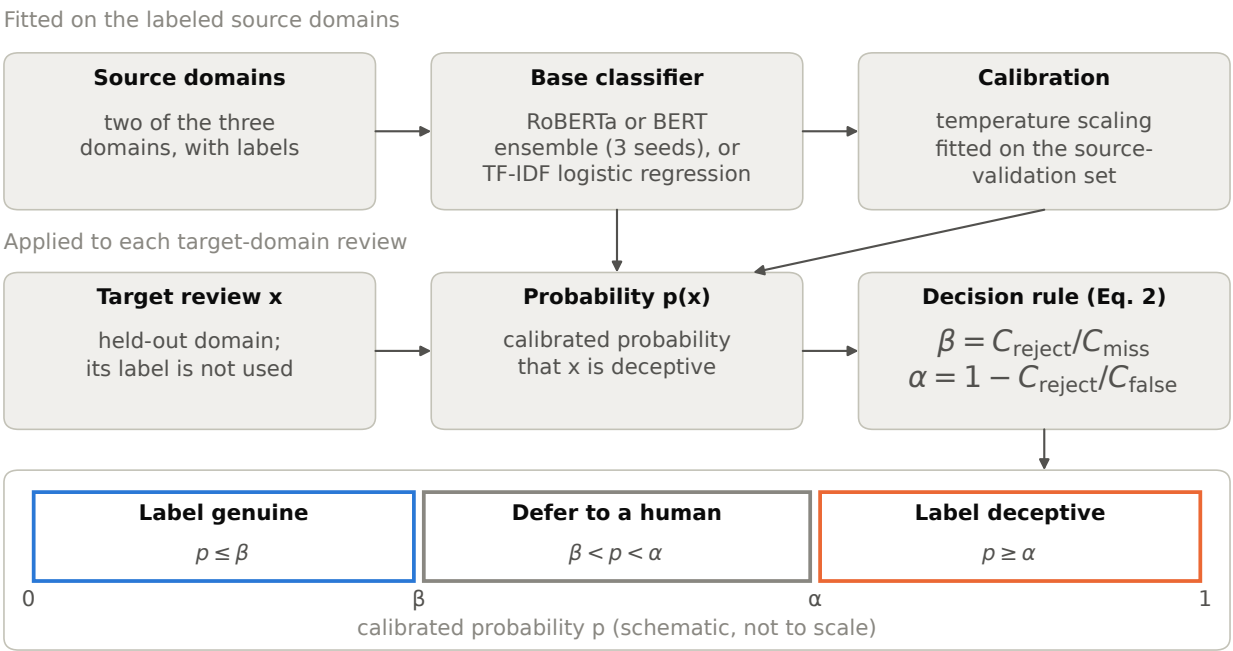
Fitted on the labeled source domains
Source domains
two of the three domains, with labels
Base classifier
RoBERTa or BERT ensemble (3 seeds), or TF-IDF logistic regression
Calibration
temperature scaling fitted on the source-validation set
Applied to each target-domain review
Target review x
held-out domain; its label is not used
Probability p(x)
calibrated probability that x is deceptive
Decision rule (Eq. 2)
$\beta = C_{\text{reject}}/C_{\text{miss}}$
$\alpha = 1 - C_{\text{reject}}/C_{\text{false}}$
Label genuine
$p \le \beta$
Defer to a human
$\beta < p < \alpha$
Label deceptive
$p \ge \alpha$
0
β
α
1
calibrated probability p (schematic, not to scale)